THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Coordinate Geometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 12	Let the point be $(0, y)$
2	**(d)** 63	Twice the area
3	**(b)** $\frac{16}{3}$	Equate squared distances
4	**(a)** 6	$x = 0$: $mc + na = 0$
5	**(c)** 50	$s^2 = d^2/2$
6	**(d)** 28	Use the area formula
7	**(a)** 9	Let the point be $(0, y)$
8	**(b)** 44	Twice the area
9	**(c)** 4	Equate squared distances
10	**(c)** 8	$x = 0$: $mc + na = 0$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$AB^2 = 25$, $BC^2 = 25$, $CD^2 = 25$, $DA^2 = 25$ (2); all four sides equal (1); diagonals $AC^2 = 80$, $BD^2 = 20$ differ so it is a rhombus, not a square (2).	2+1+2
12	Area of triangle must be 0 (1); $\frac{1}{2}\lvert 1(8-k) + 2(k-5) + 4(5-8)\rvert = 0$ (1); $k = 14$ (1).	1+1+1
13	Midpoint of the opposite side $= (3,\ 4)$ (1); length $= 5$ (1).	1+1
14	$AB^2 = 25$, $BC^2 = 25$, $CD^2 = 25$, $DA^2 = 25$ (2); all four sides equal (1); diagonals $AC^2 = 80$, $BD^2 = 20$ differ so it is a rhombus, not a square (2).	2+1+2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$AB^2 = 25$, $BC^2 = 25$, $CD^2 = 25$, $DA^2 = 25$ (2); all four sides equal (1); diagonals $AC^2 = 90$, $BD^2 = 10$ differ so it is a rhombus, not a square (2).	2+1+2
16	P divides AB in $1:2$: $(3,\ 2)$ (1.5); Q divides AB in $2:1$: $(6,\ 4)$ (1.5).	1.5+1.5
17	$(6)^2 + (y-3)^2 = 100$ (1); $(y-3)^2 = 64$ (1); $y = 3 \pm 8$, i.e. $y = 11$ or -5 (1).	1+1+1
	OR: $AB^2 = 100$, $BC^2 = 100$, $CD^2 = 100$, $DA^2 = 100$ (2); all four sides equal (1); diagonals $AC^2 = 320$, $BD^2 = 80$ differ so it is a rhombus, not a square (2).	2+1+2
18	P divides AB in $1:2$: $(6,\ 5)$ (1.5); Q divides AB in $2:1$: $(9,\ 7)$ (1.5).	1.5+1.5
	OR: $(12)^2 + (y-1)^2 = 169$ (1); $(y-1)^2 = 25$ (1); $y = 1 \pm 5$, i.e. $y = 6$ or -4 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Centroid $= (6,\ \frac{8}{3})$ (2); area $= \frac{1}{2}\lvert \dots \rvert = 48$ sq. units (3).	2+3
	OR: Diagonals bisect each other: midpoint of AC is $(1, 2)$ (2); this is also the midpoint of BD (1); $D = (1,\ 4)$ (2).	2+1+2
20	Centroid $= (4,\ 4)$ (2); area $= \frac{1}{2}\lvert \dots \rvert = 18$ sq. units (3).	2+3
	OR: Diagonals bisect each other: midpoint of AC is $\left(\frac{11}{2}, 7\right)$ (2); this is also the midpoint of BD (1); $D = (4,\ 11)$ (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Coordinate Geometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 50	$s^2 = d^2/2$
2	**(c)** 28	Use the area formula
3	**(c)** $\frac{22}{3}$	Let the point be $(0, y)$
4	**(b)** 78	Twice the area
5	**(b)** $\frac{17}{5}$	Equate squared distances
6	**(b)** 10	$x = 0$: $mc + na = 0$
7	**(d)** 50	$s^2 = d^2/2$
8	**(d)** 30	Use the area formula
9	**(a)** 6	Let the point be $(0, y)$
10	**(b)** 120	Twice the area

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Area of triangle must be 0 (1); $\frac{1}{2}\lvert -2(7-k) + 0(k-3) + 3(3-7)\rvert = 0$ (1); $k = 13$ (1).	1+1+1
12	Midpoint of the opposite side $= (5,\ 6)$ (1); length $= 5$ (1).	1+1
13	$AB^2 = 25$, $BC^2 = 25$, $CD^2 = 25$, $DA^2 = 25$ (2); all four sides equal (1); diagonals $AC^2 = 90$, $BD^2 = 10$ differ so it is a rhombus, not a square (2).	2+1+2
14	Area of triangle must be 0 (1); $\frac{1}{2}\lvert 2(1-k) + 4(k--3) + 5(-3-1)\rvert = 0$ (1); $k = 3$ (1).	1+1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$AB^2 = 25$, $BC^2 = 25$, $CD^2 = 25$, $DA^2 = 25$ (2); all four sides equal (1); diagonals $AC^2 = 80$, $BD^2 = 20$ differ so it is a rhombus, not a square (2).	2+1+2
16	P divides AB in $1:2$: $(0,\ 4)$ (1.5); Q divides AB in $2:1$: $(3,\ 2)$ (1.5).	1.5+1.5
17	$(8)^2 + (y-2)^2 = 100$ (1); $(y-2)^2 = 36$ (1); $y = 2 \pm 6$, i.e. $y = 8$ or -4 (1).	1+1+1
	OR: $AB^2 = 25$, $BC^2 = 25$, $CD^2 = 25$, $DA^2 = 25$ (2); all four sides equal (1); diagonals $AC^2 = 90$, $BD^2 = 10$ differ so it is a rhombus, not a square (2).	2+1+2
18	P divides AB in $1:2$: $(4,\ 1)$ (1.5); Q divides AB in $2:1$: $(7,\ 4)$ (1.5).	1.5+1.5
	OR: $(12)^2 + (y-0)^2 = 169$ (1); $(y-0)^2 = 25$ (1); $y = 0 \pm 5$, i.e. $y = 5$ or -5 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Centroid $= (6,\ 3)$ (2); area $= \frac{1}{2}\lvert \dots \rvert = 24$ sq. units (3).	2+3
	OR: Diagonals bisect each other: midpoint of AC is $\left(\frac{9}{2}, \frac{9}{2}\right)$ (2); this is also the midpoint of BD (1); $D = (3,\ 8)$ (2).	2+1+2
20	Centroid $= (6,\ 6)$ (2); area $= \frac{1}{2}\lvert \dots \rvert = 27$ sq. units (3).	2+3
	OR: Diagonals bisect each other: midpoint of AC is $(7, 2)$ (2); this is also the midpoint of BD (1); $D = (6,\ 4)$ (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Coordinate Geometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 4	Equate squared distances
2	**(d)** 12	$x = 0$: $mc + na = 0$
3	**(d)** 100	$s^2 = d^2/2$
4	**(a)** 42	Use the area formula
5	**(b)** $-\frac{9}{2}$	Let the point be $(0, y)$
6	**(d)** 60	Twice the area
7	**(a)** 10	Equate squared distances
8	**(b)** 9	$x = 0$: $mc + na = 0$
9	**(a)** 50	$s^2 = d^2/2$
10	**(b)** 50	Use the area formula

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Midpoint of the opposite side $= (4,\ 5)$ (1); length $= 5$ (1).	1+1
12	$AB^2 = 25$, $BC^2 = 25$, $CD^2 = 25$, $DA^2 = 25$ (2); all four sides equal (1); diagonals $AC^2 = 80$, $BD^2 = 20$ differ so it is a rhombus, not a square (2).	2+1+2
13	Area of triangle must be 0 (1); $\frac{1}{2}\lvert 3(8-k) + 6(k-2) + 9(2-8)\rvert = 0$ (1); $k = 14$ (1).	1+1+1
14	Midpoint of the opposite side $= (6,\ 4)$ (1); length $= 5$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$AB^2 = 25$, $BC^2 = 25$, $CD^2 = 25$, $DA^2 = 25$ (2); all four sides equal (1); diagonals $AC^2 = 80$, $BD^2 = 20$ differ so it is a rhombus, not a square (2).	2+1+2
16	P divides AB in $1:2$: $(5,\ 3)$ (1.5); Q divides AB in $2:1$: $(8,\ 1)$ (1.5).	1.5+1.5
17	$(8)^2 + (y-2)^2 = 100$ (1); $(y-2)^2 = 36$ (1); $y = 2 \pm 6$, i.e. $y = 8$ or -4 (1).	1+1+1
	OR: $AB^2 = 25$, $BC^2 = 25$, $CD^2 = 25$, $DA^2 = 25$ (2); all four sides equal (1); diagonals $AC^2 = 80$, $BD^2 = 20$ differ so it is a rhombus, not a square (2).	2+1+2
18	P divides AB in $1:2$: $(4,\ 5)$ (1.5); Q divides AB in $2:1$: $(8,\ 7)$ (1.5).	1.5+1.5
	OR: $(8)^2 + (y--1)^2 = 289$ (1); $(y--1)^2 = 225$ (1); $y = -1 \pm 15$, i.e. $y = 14$ or -16 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Centroid $= (4,\ \frac{11}{3})$ (2); area $= \frac{1}{2}\lvert \ldots \rvert = 32$ sq. units (3).	2+3
	OR: Diagonals bisect each other: midpoint of AC is $\left(\frac{1}{2}, 2\right)$ (2); this is also the midpoint of BD (1); $D = (-1,\ 6)$ (2).	2+1+2
20	Centroid $= (1,\ 2)$ (2); area $= \frac{1}{2}\lvert \ldots \rvert = 18$ sq. units (3).	2+3
	OR: Diagonals bisect each other: midpoint of AC is $(7, 8)$ (2); this is also the midpoint of BD (1); $D = (5,\ 12)$ (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.